Research Highlights

- Four packet-level attacker actions are modeled as Lie-cone generators on flows.
- Trajectories in the cone provably preserve rate, length, header and MTU invariants.
- FFE matches PSO (59% vs 54%) at 7,500× lower cost and half the queries.
- Physical feasibility reaches 100% while mathematical baselines collapse to 0–1%.
- Dual defense (MPS + LGAT) slashes FFE success from 51.6% to 5.3% at M=5.